SOLUTIONS: DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

1. a) Let $\Delta(s)$ and $H(s)$ be stable. The feedback loop is stable if $\|\Delta H\|_\infty < 1$.

b) For SISO systems, $\|\Delta H\|_\infty < 1$ is equivalent to the Nyquist diagram of $\Delta H$ being strictly inside the unit circle centered around the origin. It follows that there are no encirclements of the point 1 and since $\Delta$ and $H$ are stable, it follows that the closed–loop is stable.

c) Taking the first hint, we can express $\varepsilon$ in terms of $\delta$ from Figure 1 in the question as follows:

$$\varepsilon = \delta_1 + K(\delta_2 + G\varepsilon)$$

or

$$\varepsilon = (I-KG)^{-1}(\delta_1 + K\delta_2) = \overbrace{(I-KG)^{-1}\begin{bmatrix} I & K \end{bmatrix}}^{H}\begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix}$$

It follows that we can use the small gain theorem to conclude that a sufficient condition for robust stability is that $H$ is stable and $\|H\|_\infty < 1$ since $\Delta$ is stable and $\|\Delta\|_\infty < 1$.

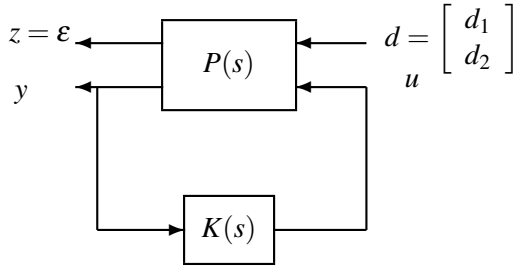
d) Taking the second hint, we can express $\varepsilon$ in terms of $d$ from Figure 1 in the question as follows:

$$\varepsilon = d_1 + K(d_2 + G\varepsilon)$$

or

$$\varepsilon = (I-KG)^{-1}(d_1 + Kd_2) = \overbrace{(I-KG)^{-1}\begin{bmatrix} I & K \end{bmatrix}}^{H}\begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

It follows from part (c) that the robust stability design specifications reduce to the requirement that the transfer matrix from $d$ to $z = \varepsilon$ in the following generalized regulator has $\mathscr{H}_\infty$-norm less than 1. The generalized regulator problem: find an internally stabilizing $K$ such that $\|T_{zd}\|_\infty < 1$, where $T_{zd}$ is the transfer matrix from $d$ to $z$ and $P(s)$ is given below.




$$P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} = \left[\begin{array}{cc|c} I & 0 & I \\ \hline G(s) & I & G(s) \end{array}\right].$$

2. a) Assume first that $A$ is stable. Then $\|H\|_\infty < \gamma$ if and only if, with $x(0) = 0$, $J := \int_0^\infty [y^T y - \gamma^2 u^T u]dt < 0$, for all $u(t)$ such that $\|u\|_2 < \infty$. If $\|u\|_2$ is bounded, then $\lim_{t\to\infty} x(t) = 0$. Now, $\int_0^\infty \frac{\mathrm{d}}{\mathrm{dt}}[x^T Px]dt = x(\infty)^T Px(\infty) - x(0)^T Px(0) = 0$. So,

$$0 = \int_0^\infty \left(\dot{x}^T Px + x^T P\dot{x}\right) dt = \int_0^\infty [x^T (A^T P + PA)x + x^T PBu + u^T B^T Px]dt.$$

Use $y = Cx + Du$ and add the last expression to $J$

$$\begin{aligned} J &= \int_0^\infty [x^T (A^T P + PA + C^T C)x + 2x^T (PB + C^T D)u + u^T (D^T D - \gamma^2 I)u]dt \\ &= \int_0^\infty \begin{bmatrix} x^T & u^T \end{bmatrix} \overbrace{\begin{bmatrix} A^T P + PA + C^T C & PB + C^T D \\ B^T P + D^T C & D^T D - \gamma^2 I \end{bmatrix}}^{M} \begin{bmatrix} x \\ u \end{bmatrix} dt. \end{aligned}$$

It follows that $J < 0$, and so $\|H\|_\infty < \gamma$, if $P \succ 0$ and $M \prec 0$. It remains to prove the stability of $A$. Note that from the (1,1) entry of $M$ we have $A^T P + PA \prec 0$. Let $Az = \lambda z$, $z \neq 0$. Then multiplying the inequality from the left by $z^T$ and from the right by $z$ gives $(\lambda + \bar{\lambda})z^T Pz < 0$. Since $P \succ 0$ it follows that $z^T Pz > 0$ and it follows that $\lambda + \bar{\lambda} < 0$ so that $A$ is stable.

b) i) The state space equations are

$$\begin{aligned} \dot{x}(t) &= Ax(t) + BK(r(t) - Cx(t)) = (A - BKC)x(t) + BKr(t) \\ y(t) &= Cx(t) \end{aligned}$$

Therefore $T_{yr}(s) = C\left(sI - (A - BKC)\right)^{-1} BK$.

ii) Using the answer to Part a, replacing $A$ by $A - BKC$, $B$ by $BK$ and setting $D = 0$, there exists a feasible $K$ if there exists $P \succ 0$ such that

$$\begin{bmatrix} (A - BKC)^T P + P(A - BKC) + C^T C & PBK \\ K^T B^T P & -\gamma^2 I \end{bmatrix} \prec 0.$$

iii) The only nonlinearity is due to the product $PBK$. Using the suggested variable defined via $PB = BX$ (which is a linear equation in the variables $P$ and $X$) the matrix inequality becomes

$$\begin{bmatrix} A^T P + PA - B(XK)C - C^T (XK)^T B^T + C^T C & B(XK) \\ (XK)^T B^T & -\gamma^2 I \end{bmatrix} \prec 0.$$

This then becomes linear by defining the variable $\hat{K} = XK$. Finally, $K$ can be recovered from $\hat{K}$ provided that $X$ is nonsingular.

3. a) It is assumed that the triple $(A,B,C)$ is minimal. A suitable Lyapunov function for regulating $x$ is $V = x^T Px$ where $P = P^T$. Set $u = -Fx$. Provided that $P = P^T \succ 0$ and $\dot{V} < 0$ along closed-loop trajectories, we can assume $\lim_{t\to\infty} x(t) = 0$. Then $\dot{V} = \dot{x}^T Px + x^T P\dot{x} = x^T \left(A^T P + PA - F^T B^T P - PBF\right) x$. Using $x(\infty) = 0$, $\int_0^\infty \dot{V}(t)dt = \int_0^\infty x^T \left(A^T P + PA - F^T B^T P - PBF\right) xdt = -x_0^T Px_0$.

b) Adding the last equation to the expression for $J$ and completing a square:

$$J = x_0^T Px_0 + \int_0^\infty \{x^T [A^T P + PA + C^T C - PBB^T P]x + \left\|(F - B^T P)x\right\|^2\} dt.$$

Since the last term is always nonnegative, it follows that the minimizing value of $F$ is given by $F = B^T P$. We can set the term in square brackets to zero provided $P$ satisfies the Riccati equation $A^T P + PA + C^T C - PBB^T P = 0$. It follows that the minimum value of $J$ is $x_0^T Px_0$.

c) We prove that $A_c := A - BB'P$ is stable. The Riccati equation can be written as

$$A_c' P + PA_c + C^T C + PBB'P = 0.$$

Let $\lambda \in \mathscr{C}$ be an eigenvalue of $A_c$ and $y \neq 0$ be the corresponding eigenvector. Pre– and post–multiplying the Riccati equation be $y'$ and $y$ respectively gives $(\lambda + \bar{\lambda})y'Py + y'C'Cy + y'PBB'Py = 0$. Since $P \succ 0$ and $y \neq 0$, $y'Py > 0$, $y'C'Cy \geq 0$ and $y'PBB'Py \geq 0$. It follows that $\lambda + \bar{\lambda} \leq 0$ with $\lambda + \bar{\lambda} = 0$ if and only if $B'Py = 0$ and $Cy = 0$. But this implies $Ay = \lambda y$ and $Cy = 0$, which contradicts the observability of the pair $(A,C)$. This proves that $\lambda + \bar{\lambda} < 0$ and so the closed-loop is stable.

d) Now, $\dot{x} = Ax + Bu + Bd = Ax + B(d - Fx)$. It follows that

$$\begin{aligned} \dot{x} &= (A - BF)x + Bd \\ z_1 &= Cx \end{aligned}$$

it follows that $T_{z_1 d} \stackrel{s}{=} (A - BF, B, C, 0)$. It follows from the bounded real lemma that $||T_{z_1 d}||_\infty < \gamma$ if there exists $P = P^T \succ 0$ such that

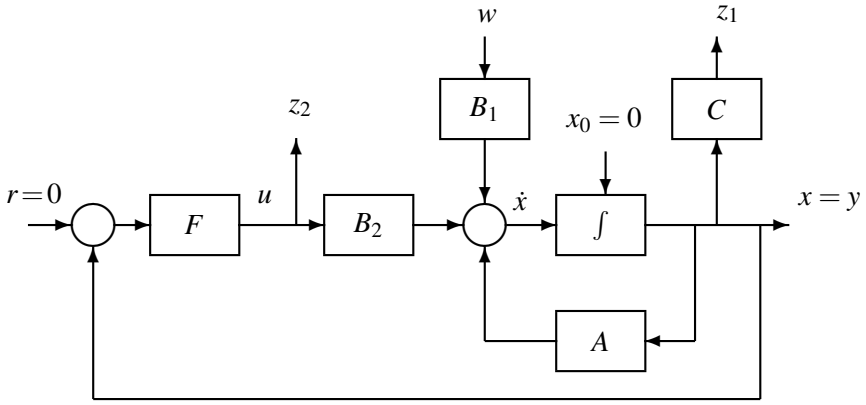
$$\begin{bmatrix} P(A - BF) + (A - BF)^T P + C^T C & PB \\ B^T P & -\gamma^2 I \end{bmatrix} \prec 0$$

Using a Schur complement argument, this inequality is equivalent to

$$P(A - BF) + (A - BF)^T P + C^T C + \gamma^{-2} PBB^T P \prec 0.$$

However, it follows from the Riccati equation in Part c above that $P(A - BF) + (A - BF)^T P + C^T C + PBB^T P = 0$. Ignoring issues relating to strict matrix inequalities, this proves that $||T_{z_1 d}||_\infty \leq 1$ and so 1 is an upper bound on $||T_{z_1 d}||_\infty$.

4. a) The block diagram is given as




b) Let $J = \|z\|_2^2 - \gamma^2\|w\|_2^2$ denote the cost function. Let $V = x'Xx$ and set $u = Fx$. Provided that $X = X' \succ 0$ and $\dot{V} < 0$ along the closed-loop trajectory, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Xx + x'X\dot{x} = x'\left(A'X + XA + F'B_2'X + XB_2F\right)x + x'XB_1w + w'B_1'Xx.$$

Integrating from 0 to $\infty$ and using $x(\infty) = x_0 = 0$,

$$0 = \int_0^\infty [x'\left(A'X + XA + F'B_2'X + XB_2F\right)x + x'XB_1w + w'B_1'Xx]dt.$$

Adding the last equation to the definition of $J$ and completing squares

$$J = \int_0^\infty \{x'\left[A'X + XA + C'C - X\left(B_2B_2' - \gamma^{-2}B_1B_1'\right)X\right]x + \|Zx\|^2 - \|v\|^2\}dt$$

where $Z = F + B_2'X$ and $v = \gamma w - \gamma^{-1}B_1'Xx$.

c) The requirement $\|H\|_\infty \le \gamma$ is equivalent to $J \le 0$. Thus sufficient conditions for $J \le 0$ are $Z = 0$ and the existence of $X$ s.t.

$$A'X + XA + C'C - X\left(B_2B_2' - \gamma^{-2}B_1B_1'\right)X = 0, \qquad X = X' \succ 0.$$

Setting $Z = 0$, the control policy is $F = -B_2'X$. Setting $v = 0$, the worst case disturbance is $w^* = \gamma^{-2}B_1'Xx$. The closed-loop with $u = Fx$ and $w = w^*$ is $\dot{x} = A_cx$ where $A_c = A - \left(B_2B_2' - \gamma^{-2}B_1B_1'\right)X$ and a further condition is $Re\ \lambda_i(A_c) < 0,\ \forall\, i$.

d) Setting $B_1 = B_2 = B$ and using the solution in Part c) with $\gamma = 1$, it follows that the Riccati equation becomes

$$A'X + XA + C'C = 0, \qquad X = X' \succ 0.$$

Thus, since $A_c = A$, it follows that a condition for the solution of the problem is that $A$ is stable.